

MTL1002: TUTORIAL 5
(DIFFERENTIAL EQUATIONS)

- (1) Find the solutions of the following initial value problems:

(a) $y'' - 2y' - 3y = 0, \quad y(0) = 0, \quad y'(0) = 1$
(b) $y'' + 10y = 0, \quad y(0) = \pi, \quad y'(0) = \pi^2$

- (2) Find a function ϕ which has a continuous derivative on $0 \leq x \leq 2$ which satisfies $\phi(0) = 0, \phi'(0) = 1$ together with

$$\begin{aligned} y'' - y &= 0 & \text{for } 0 \leq x \leq 1 & \text{ and,} \\ y'' - 9y &= 0 & \text{for } 1 \leq x \leq 2. \end{aligned}$$

- (3) Consider the following ODE with constant coefficient: $L(y) = y'' + a_1y' + a_2y = 0$. Let ϕ_1 and ϕ_2 be the solutions satisfying

$$\begin{aligned} \phi_1(x_0) &= 1, & \phi_1'(x_0) &= 0, & \text{and} \\ \phi_2(x_0) &= 0, & \phi_2'(x_0) &= 1. \end{aligned}$$

Assume that ϕ is a solution satisfying

$$\phi(x_0) = \alpha, \quad \phi'(x_0) = \beta.$$

Prove that $\phi(x) = \alpha\phi_1(x) + \beta\phi_2(x)$ for all x .

- (4) Let ϕ_1, ϕ_2 be two differentiable functions on an interval I , which are not necessarily solutions of an equation $L(y) = 0$. Prove the following statements.

- (a) If ϕ_1, ϕ_2 are linearly dependent on I , then $W(\phi_1, \phi_2)(x) = 0$ for all x in I .
(b) If $W(\phi_1, \phi_2)(x_0) \neq 0$ for some x_0 in I , then ϕ_1, ϕ_2 are linearly independent on I .
(c) $W(\phi_1, \phi_2)(x) = 0$ for all x in I does not imply that ϕ_1, ϕ_2 are linearly dependent on I .
(d) $W(\phi_1, \phi_2)(x) = 0$ for all x in I , and $\phi_2(x) \neq 0$ on I , imply that ϕ_1, ϕ_2 are linearly dependent on I .

- (5) Find all solutions of the following ODEs:

- (a) $4y'' - y = e^x$
(b) $y'' + 4y = \cos x$
(c) $y'' + 9y = \sin 3x$

- (6) Let $L(y) = y'' + a_1y' + a_2y = 0$, where a_1, a_2 are constants, and let the characteristic equation be $p(r) = r^2 + a_1r + a_2 = 0$.

- (a) If A, α are constants and $p(\alpha) \neq 0$, show that there is a solution ϕ of $L(y) = Ae^{\alpha x}$ of the form $\phi(x) = Be^{\alpha x}$, where B is a constant.

- (b) Compute a particular solution of $L(y) = Ae^{\alpha x}$ in case $p(\alpha) = 0$.

- (7) Are the following set of functions defined on $-\infty < x < \infty$ linearly dependent or independent? Why?
- (a) $\phi_1(x) = 1, \quad \phi_2(x) = x, \quad \phi_3(x) = x^3$
- (b) $\phi_1(x) = x, \quad \phi_2(x) = e^{2x}, \quad \phi_3(x) = |x|$
- (8) Use the method of undetermined coefficients to find a particular solution of each of the following equations:
- (a) $y'' + 4y = \cos x$
- (b) $y'' + 4y = \sin 2x$
- (c) $y'' - y' - 2y = x^2 + \cos x$
- (d) $y'' + 9y = x^2 e^{3x}$.
- (9) Find a real solution:
- (a) $x^2 y'' - 4xy' + 6y = 0$
- (b) $4x^2 y'' + 12xy' + 3y = 0$
- (c) $x^2 y'' + 7xy' + 9y = 0$
- (d) $x^2 y'' - 2.5xy' - 2y = 0$
- (e) $x^2 y'' + 7xy' + 13y = 0$
- (10) Solve the following initial value problems:
- (a) $x^2 y'' - 2xy' + 2y = 0, y(1) = 1.5, y'(1) = 1$
- (b) $x^2 y'' + 3xy' + y = 0, y(1) = 3, y'(1) = -4$
- (c) $x^2 y'' - 3xy' + 4y = 0, y(1) = 0, y'(1) = 3$
- (11) Find all solutions of the following ODEs:
- (a) $y''' - 8y = 0$
- (b) $y^{(4)} + 16y = 0$
- (c) $y^{(100)} + 100y = 0$
- (d) $y^{(4)} - 16y = 0$
- (12) Use the variation of parameters method to solve the following ODEs:
- (a) $y''' - y' = x$
- (b) $y^{(4)} + 16y = \cos x$
- (c) $y^{(4)} - 4y^{(3)} + 6y'' - 4y' + y = e^x$